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Career Objective

To pursue a career in the field of technology by keeping up pace with the latest trends and working dynamically for the growth of organization. In short, seeking a position, where I can get hands-on experience on the state-of-the-art techniques for industrial-strength designs and exposure towards incorporating new research directions into the industrial context.

Professional Experience

- October, 2024 – Present **Assistant Professor** at the Department of Computer Science and Engineering (CSE), C. V. Raman Global University, Bhubaneswar, Odisha, India.
- September, 2021–July, 2024 **Research and Engineering Scientist** at Zoho Corporation, Chennai, Tamil Nadu, India.
- May, 2018–October, 2020 **Post Doctoral Fellow (PDF)** at the Department of Electrical Engineering and Computer Science (EECS), Indian Institute of Science Education and Research (IISER), Bhopal, Madhya Pradesh, India.

Research Areas

- October 2024–Present My current research focuses on the applied inter-disciplinary areas of **Software Requirements Engineering**, **Computational Linguistics**, **Natural Language Processing** and **Deep Learning**.
- 2021–2024 **Programming Language Design and Implementation (PLDI)**: The main duties and responsibilities are to work with the engineering teams in developing technological innovations for products, independently drafting technical specifications describing the system design and technology, interacting with the engineering teams to identify potential areas of innovation and engaging in independent research and study to learn technology as required.
- 2018–2020 The intersected areas of **Software Requirements Engineering** (User Stories & SRS Specifications), **Natural Language Processing** (NLP), **Word Embeddings** (Word2Vec, FastText, Glove & BERT) and **Machine Learning**: Related to these aspects, we have designed and developed efficient NLP and word embedding based approaches that are helpful to extract ambiguous words (two different meaning for the same word), glossaries (key terms) and to cluster the extracted glossaries for large-sized natural language documents. The above mentioned used approaches are used to improve the quality of the software requirements written in natural language.
- Past Research The sub-domain areas of software engineering including the **Software Service Effort Estimation**, **Testing** of Service-Oriented Architecture (SOA) Systems and **Program Analysis** of Aspect-Oriented Programs (AOP) have been explored as a part of my past research during my doctoral and masters study.

Education

- 2018–2020 **Post Doctoral Fellow (PDF)**, Department of Electrical Engineering and Computer Science (EECS), Indian Institute of Science Education and Research (IISER), Bhopal, Madhya Pradesh, India
- 2013–2018 **Ph.D. in Computer Science and Engineering**, Indian Institute of Technology (Indian School of Mines), IIT (ISM), Dhanbad, Jharkhand, India
- 2010–2012 **Master of Technology (M. Tech) in Computer Science and Engineering**, Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar, Odisha, India

- 2005–2009 **Bachelor of Technology (B. Tech) in Computer Science and Engineering**, *Biju Patnaik University of Technology (BPUT)*, Rourkela, Odisha, India
- 2005 **Higher Secondary**, *Kendriya Vidyalaya, Puri*, Odisha, India
- 2003 **Secondary**, *Kendriya Vidyalaya, Puri*, Odisha, India

Doctoral Thesis

- Title *Efficient Cost Estimation and Testing Approaches for SOA Systems*
- Supervisor Prof. Chiranjeev Kumar, Department of Computer Science and Engineering, IIT (ISM), Dhanbad, Jharkhand, India
- Short Thesis Description The main objective is to study, design and develop efficient cost estimation and testing approaches for SOA systems. In general, the cost/effort estimation and testing of SOA systems is different from that of traditional software. One of the prominent reason for this could be visualized from the architectural difference of SOA systems when compared to the traditional software development. Additionally, some recent empirical studies pointed out that distributed processing is more complicated than the non-distributed processing. As any form loose coupling make tasks easier, but increases complexity. In the literature, there is a lack of significant work about cost and effort estimation of SOA systems. Addressing the above mentioned issues, two novel effort estimation approaches have been developed by utilizing statistics and machine learning. The first approach deals in utilizing statistical based approach in estimating the development effort of SOA systems from the implication of new proposed metrics and regression models based on some past project data. The second approach deals in employing machine learning based technique. The proposed structure employs an adaptive filter by combining the insights of signals theory with empirical research to find a recommendation scheme for SOA system effort estimation. To the best of our knowledge, this is the first work in literature to solve the problem by adhering basic principles of SOA system design with a proper validation mean. Furthermore, the current testing approaches of SOA systems are inadequate in addressing the complexities associated with the SOA systems. Additionally, the current testing techniques of SOA systems are unable to assure the desired level of trust, which may create a barrier between the service providers and the end users. Therefore, we inspect an abstract formulation of this problem to identify faulty business processes using minimum number of tests for improving the testing effort and fostering trust between the service providers and end users.

Master Thesis

- Title *Dynamic Slicing of Concurrent Aspect-oriented Programs*
- Supervisor Dr. Abhishek Ray (Associate Professor), School of Computer Engineering, Kiit University, Bhubaneswar, Odisha, India
- Short Thesis Description The main *idea* is to deal with *dynamic slicing of concurrent aspect-oriented programs*. In this regard, a dependence based intermediate program representation known as Concurrent Aspect-oriented System Dependence Graph (CASDG) has been proposed to represent concurrent aspect-oriented programs. Moreover, an efficient *dynamic slicing algorithm* has been proposed and implemented for computing the *dynamic slices* of the concurrent aspect-oriented programs, appropriately during the runtime. The CASDG is modeled by the main program together with all the *methods*, *constructors*, *used* and *defined* variables present in the *non-aspect code* and all the *pointcuts*, *advices* and *introductions* exist in the *aspect code*. The proposed dynamic slicing algorithm is an extended version of Node Marking Dynamic Slicing (NMDS) algorithm for concurrent object-oriented programs, which is primarily based on *marking* and *un-marking* of the executed nodes in the CASDG.

Achievements, Awards and Invited Services

- October 2nd, 2026 **Expert Talk**, *Unlocking Prompt Engineering from Tokens to RAG*, organized by IEEE Computer Society Chapter, Bhubaneswar Section, India
- 14th–23rd September, 2026 **Resource Person**, *Online E & ICT Faculty Development Programme (FDP) on Large Language Models for Generative AI: Foundations, Design and Applications (LLMGAI)*, organized by NIT Warangal, Telangana
- December, 2024 **Invited Speaker**, *Online E & ICT Faculty Development Programme (FDP) on Real World AI Applications: Bridging Academia and Industry*, organized by NIT Warangal, Telangana in association with RNS Institute of Technology, Bengaluru

- March, 2021 **Invited Speaker**, *FDP on Scientific Writing using Latex*, organized by MITS, Madanapalle, Andhra Pradesh, India
- October, 2020 **Invited Speaker**, *FDP on Scientific Writing using Latex*, organized by BBGSU, Rajouri, J&K, India
- May, 2018 - **Institute Post Doctoral Fellowship**, *selected in merit to avail full time institute post doctoral fellowship at the Department of Electrical Engineering and Computer Science (EECS), Indian Institute of Science Engineering and Research (IISER), Bhopal, Madhya Pradesh, India*
- Present
- May, 2018 **My PhD Thesis is indexed in Association for Computing Machinery's Special Interest Group on Software Engineering (ACM SIGSOFT)**, *list of PhD Dissertations*, in the area of Software Engineering
- March, 2018 **Invited Speaker**, *Scientific Writing using Latex - A Technical Development Program (TDP)*, sponsored by a student chapter, The International Society for Optics and Photonics (SPIE) organized by, Department of Electronics Engineering, IIT (ISM), Dhanbad
- 19th March, 2018 **Successfully Defended PhD Thesis**
- December, 2017 **Winter School in Software Engineering (WSSE'2017)**, *Selected in Merit to attend Winter School in Software Engineering out of 350 participants (22/350)*, organized by Tata Research Development and Design Centre (TRDDC), Pune, Maharashtra, India and sponsored by Association for Computing Machinery - India SOFTWARE Engineering community (ACM-ISOFTE)
- 2014-2018 **Ph.D. Student Representative**, *Computer Science and Engineering Society (CSES Society), Department of Computer Science and Engineering, IIT(ISM), Dhanbad*
- 2013-2018 **ISM - JRF Fellowship**, *selected in merit to avail Scholarship, funded by Government of India Government of India (GOI), Ministry of Human Resource and Development (MHRD)*, on the basis of all India written examination and interview conducted by IIT (ISM), Dhanbad, Jharkhand, India
- 2005 **AISSCE - 2005**, *secured highest mark (89/100) in biology*, out of 39 students, of Kendriya Vidyalaya, Puri, Odisha, India
- 2004 **House Captain**, *lead Raman House (a Student Activity) with the position of house captain*, at Kendriya Vidyalaya, Puri, Odisha, India
- 2003 **KVS Regional Science Exhibition**, *honored with certificate of successful participation*, held at Kendriya Vidyalaya, No. 3, Mancheswar, Bhubaneswar, Odisha, India
- 2003 **National Trekking and Adventure Camp**, *participated and successfully completed the trekking and adventure camp organized by Directorate of Mountaineering and Allied Sports Manali, Government of Himachal Pradesh*, held at Himachal Pradesh, India
- 2003 **KVS Regional Level Games and Sports Meet**, *participated and bagged runners up position in men's field hockey*, held at Kendriya Vidyalaya, No. 1, Bhubaneswar, Odisha, India
- 2002 **Fifth National Science Olympiad**, *honored with certificate of successful participation*, qualified first round (level 1), held on 30th January, 2003 at more than 2050 centres in India and Gulf
- 2001 **The Green Olympiad**, *honored with certificate of participation*, sponsored by the ministry of Environment and Forests, Government of India (GOI)

Professional Activities/Service

- October, 2026 **Program Committee Member**, *26th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2027)*, Hanoi, Vietnam
- September, 2026 **External Reviewer**, *International Conference on Artificial Intelligence, Computer, Data Sciences and Applications, 2026*. Rio de Janeiro, Brazil
- June, 2026 **Program Committee Member**, *Agentic Software Engineering (SE 3.0)*, co-located with ACM KDD, 2026. Jeju Islands, South Korea
- January, 2026 **Reviewer**, *International Conference on Frontiers in Advanced Computing and Emerging Intelligent Technologies (FACEIT-2026)*, NIT Warangal, India
- August, 2025 **Reviewer**, *Nature Scientific Reports*
- August, 2025 **Invited Reviewer**, *International Conference on Cyber Security and Artificial Intelligence (ICCSAI'25)*, Kolkata, India

August, 2025 **Invited Reviewer**, *International Conference on Artificial Intelligence and Emerging Technologies (ICAJET'25)*, XIM University, Bhubaneswar, India

July, 2025 **Invited Reviewer**, *1st International Conference on Power and Intelligent Control Systems (PICS'25)*, NIT Hamirpur, India

March, 2025 **Invited Reviewer**, *6th International Conference on Recent Advances in Information Technology (RAIT'25)*, IIT (ISM), Dhanbad, India

February, 2025 **Invited Reviewer**, *1st International Conference on Smart and Sustainable Developments in Electrical Engineering (SSDEE'25)*, IIT (ISM), Dhanbad, India

February, 2025 **Reviewer**, *2nd International Conference on Artificial Intelligence, Machine Learning & Intelligent Systems (ICAMS'25)*, NIT Hamirpur, Himachal Pradesh, India

January, 2025 **Reviewer**, *6th International Conference on Advances in Distributed Computing and Machine Learning (ICADCML'25)*, National Taiwan University of Science and Technology (NTUST), Taiwan

November, 2024 **Technical and Advisory Committee**, *2nd International Conference on Artificial Intelligence, Machine Learning & Intelligent Systems (ICAMS'25)*, NIT Hamirpur, Himachal Pradesh, India

September, 2024 **Session Chair**, *International Conference on Computing, Communication and Learning (COCOLE'24)*, NIT Warangal, Telangana, India

March, 2024 **Reviewer**, *International Conference on Cognitive, Green and Ubiquitous Computing (ICCGU'24)*, CV Raman Global University, Bhubaneswar, India

July, 2023 **Sub-Reviewer**, *20th India Council International Conference (INDICON'23)*, Warangal, India

August, 2022 **Sub-Reviewer**, *3rd International Conference on Advances in Distributed Computing and Machine Learning (ICADCML'22)*, NIT Warangal, India

August, 2022 **Publicity Chair**, *International Conference on Computing, Communication and Learning (COCOLE'22)*, NIT Warangal, Telangana, India

August, 2022 **Reviewer**, *Automation in Construction*, Elsevier

June, 2022 **Reviewer**, *Frontiers in Psychology*, (section Health Psychology)

August, 2021 **Reviewer**, *Clinical Health*, Elsevier

July, 2021 **Reviewer**, *Expert System with Applications*, Elsevier

March, 2021 **Publicity Co-Chair**, *15th Innovations in Software Engineering Conference (ISEC 2022)*, DA-IICT Gandhinagar, Gujarat, India

February, 2021 **Sub-Reviewer**, *International Conference on VLSI & Microwave and Wireless Technologies (ICVMWT'21)*, Allahabad University, India., Prayagraj, India

January, 2021 **Reviewer**, *Information Fusion*, Elsevier

January, 2021 **Reviewer**, *Journal of Supercomputing*, Springer

January, 2020 **Volunteer**, *The Fourth Paradigm : From Data to Discovery*, Bhopal, India

February, 2019 **Reviewer**, *Future Generation Computer Systems (FGCS)*, Elsevier

December, 2018 **Reviewer**, *Arabian Journal for Science and Engineering (AJSE)*, Springer

November, 2018 **Reviewer**, *Journal of Parallel and Distributed Computing (JPDC)*, Elsevier

February, 2018 **Ad-Hoc Reviewer**, *International Journal of Fog Computing (IJFC)*, IGI Global

January, 2017 **Ad-Hoc Reviewer**, *International Journal of Web Services Research (IJWSR)*, IGI Global

December, 2016 **Reviewer**, *Software: Practice and Experience (SPE)*, Wiley

- November, 2016 **Guest Editor**, *Special Issue On: Scalable Computing for Knowledge Discovery, International Journal of Knowledge Discovery on Bioinformatics (IJKDB)*, IGI Global
- June, 2016 **Invited Reviewer**, *International Journal of Computers and Applications*, Taylor & Francis
- August, 2016 **Invited Reviewer**, *International Journal of Rough Sets and Data Analysis (IJRSDA)*, IGI Global
- March, 2016 **Student Volunteer**, *3rd IEEE International Conference on Recent Advances in Information Technology (RAIT)*, Dhanbad, Jharkhand, India
- December, 2015 **Invited Reviewer**, *3rd IEEE International Conference on Computing for Sustainable Global Development (INDIACom)*, New Delhi, India
- June, 2014 **External Reviewer**, *2nd IEEE International Conference on Systems and Informatics (ICSAI)*, Shanghai, China
- March, 2014 **Student Volunteer**, *2nd International Conference on Recent Advances in Information Technology (RAIT)*, Dhanbad, India
- February, 2011 **Student Volunteer**, *7th International Conference on Distributed Computing and Information Technology (ICDCIT)*, Bhubaneswar, India

Skill Sets

- Languages: **C, C++, R, HTML, Core Java, Swift, Python and AspectJ**
- Operating Systems: **Windows and Linux**
- Model: **NuSMV and PRISM**
- Checkers:
- Database: **MySQL**
- Packages: **Microsoft Office and LibreOffice**
- Typesetting System: **L^AT_EX**
- Scripting Languages: **Java Script and VB Script**
- Applications: **Spreadsheet, IBM Rational Software Architect, Matlab**
- Visualization Tools: **SPSS, Corel Draw, Origin, Edraw Max, Dia, yEd, ORA and Pajek**
- Word Embeddings: **Word2vec, FastText, BERT, Glove**

Teaching Assistantships

- ECS202 **Data Structures and Algorithms**, BS-MS 2nd Year (Winter 2020) at IISER Bhopal.
- ECS307 **Theory of Computation**, BS-MS 3rd Year (Monsoon 2019) at IISER Bhopal.
- ECS302 **Introduction to Software Modeling and Verification**, (Lab Session) using New Symbolic Model Verifier (NuSMV) at IISER Bhopal.
- CSC16205 **Database Management Systems Lab**, B. Tech 6th Semester, (Winter 2016-17) at IIT (ISM), Dhanbad.
- CSE11301 **C Programming Lab**, B. Tech 1st Semester, (Monsoon 2013, Winter 2015 and 2016) at IIT (ISM), Dhanbad.
- CSC15204 **Software Engineering Lab**, B. Tech 5th Semester, (Monsoon 2014, Monsoon 2015 and 7th Semester, Monsoon 2016) at IIT (ISM), Dhanbad.
- CSC14202 **Algorithm Design and Analysis Lab**, B. Tech 4th Semester, (Winter 2014) at IIT (ISM), Dhanbad.
- CSC13202 **Data Structures Lab**, B. Tech 3rd Semester, (Monsoon 2013), at IIT (ISM), Dhanbad.

Professional Affiliations

- 2025 – Present **Senior Member**, *Institute of Electrical and Electronics Engineers (IEEE)*, Member (#): 93253963

- 2020 – **Professional Member**, *Association for Computing Machinery (ACM)*, Member (#): 7254589
Present
- 2016 – **Graduate Student Member**, *Institute of Electrical and Electronics Engineers (IEEE)*, Member (#): 93253963
Present
- 2016 – **Student Member**, *Advanced Computing & Communication Society (ACCS)*, Member (#): S1688A1623481
Present
- 2016 – **Member**, *Internet Society (ISOC)*
Present
- 2015 – **Member**, *India SOFTWARE Engineering community (ISOFT)*
Present
- 2015 – 2020 **Student Member**, *Association for Computing Machinery (ACM)*, Member (#): 7254589
- 2015 – **Member**, *International Association of Engineers (IAENG)*, Member (#): 158661
Present
- 2015 – **Member**, *Computer Science Teachers Association (CSTA)*
Present
- 2015 – 2018 **Student Member**, *Universal Association of Computer and Electronics Engineers (UACEE)*, Member (#): SM10100053768
- 2014 – 2017 **Member**, *Computer Science and Engineering Society (CSE Society), ISM Dhanbad*, Member (#): 013/14-15

Workshops and Short Term Courses Attended

- 27th February, 2020 **Attended 1 day workshop co-located at the 13th Innovations in Software Engineering Conference (ISEC 2020)**, on “*Software Engineering for Artificial Intelligence*”, organized by Indian Institute of Information Technology, Design and Manufacturing (IIITDM), Jabalpur, Madhya Pradesh, India
- 11–16, December, 2017 **Attended 6 days winter school sponsored by Association for Computing Machinery, India SOFTWARE Engineering community (ACM-ISOFT)**, on “*Winter School in Software Engineering (WSSE - 2017)*”, organized by Tata Research Development and Design Centre (TRDDC), Pune, Maharashtra, India
- 09–13, February, 2017 **Attended 5 days short term course sponsored by Technical Education Quality Improvement Programme - II (TEQIP - II)**, on “*Fundamental Algorithms: Design and Analysis*”, organized by Department of Mathematics, Indian Institute of Technology, Kharagpur, West Bengal, India
- 19–23, August, 2015 **Attended 5 days short term course**, on “*Network Protocols and their Simulation using NS-2/NS-3*”, organized by Department of Computer Science & Engineering, Indian School of Mines, Dhanbad, Jharkhand, India
- 14th February, 2014 **Attended 1 day workshop**, on “*Intellectual Property Rights, under TEQIP-II Initiative*”, organized by Indian School of Mines, Dhanbad, Jharkhand, India
- 12–14, May, 2014 **Attended 3 days workshop**, on “*Recent Trends in Software Testing (RTST-2014)*”, organized by Department of Computer Science & Engineering, National Institute of Technology, Rourkela, Odisha, India
- 18–25, March, 2013 **Attended 7 days professional skill development programme**, on “*Linux Environment*”, organized by Department of Computer Science & Engineering, Indian School of Mines, Dhanbad, Jharkhand, India

Patents (Granted and Published in US Patents and Trademark Office)

1. CONSTRUCT-MODIFICATION TAGS FOR DEVELOPMENT-PHASE COMPILER REQUESTS FOR FACILITATING SOURCE CODE REVIEW AND OPTIMIZATION. **Patent Number: 12699554. (PATENT GRANT DATE: 04-08-2026).**
2. CONSTRUCT-MODIFICATION TAGS FOR DEVELOPMENT-PHASE COMPILER REQUEST. **Patent Number: 12699553. (PATENT GRANT DATE: 04-08-2026).**
3. COMPILE-TIME LINK TYPE OBJECT MANAGEMENT. **Patent Number: 12639053 (PATENT GRANT DATE: 26-05-2026).**
4. PERMISSIONS AND NOTIFICATIONS FOR CONSTRUCT-MODIFICATION TAGS. **Patent Number: 12602211 (PATENT GRANT DATE: 14-04-2026).**

5. INTEGRATED DEVELOPMENT ENVIRONMENT OBJECT MANAGEMENT CODE AUTO-SUGGESTION. **PID: 20250272063 (FILED and PUBLISHED in 2025).**
6. RUNTIME SUPPORT FOR LINK TYPE OBJECT MANAGEMENT. **PID: 20240241735 (FILED and PUBLISHED in 2024).**
7. CONSTRUCT-MODIFICATION DEACTIVATION TAG FOR DEVELOPMENT-PHASE COMPILER REQUESTS. **PID: 20240281223 (FILED and PUBLISHED in 2024).**

Publications (Peer-Reviewed Journals)

- 2026 [1] Padma Charan Sahu, Bibhu Prasad, Ratnakar Dash, Debendra Muduli, Suddhendu DasMahapatra, **Siba Mishra**, and Sourav Parija. Apr. 2026. "An Explainable Grey Wolf Optimized Extreme Learning Machine Framework for Modulation Classification in Cloud Environment". In: *Scientific Reports* 16.1, p. 19295.
- 2018 [2] Rakesh Ranjan Kumar, **Siba Mishra**, and Chiranjeev Kumar. Dec. 2018. "A Novel Framework for Cloud Service Evaluation and Selection Using Hybrid MCDM Methods". In: *Arabian Journal for Science and Engineering* 43.12, pp. 7015–7030.
- [3] **Siba Mishra** and Chiranjeev Kumar. July 2018. "Effort Estimation for Service-Oriented Computing Environments". In: *Computing and Informatics* 37.3, pp. 553–580.
- 2017 [4] Mohit Chawla, **Siba Mishra**, Kriti Singh, and Chiranjeev Kumar. Dec. 2017. "A Comparison of Data Exchange Mechanisms for Real-Time Communication". In: *International Journal of Rough Sets and Data Analysis (IJRSDA)* 4.4, pp. 66–81.
- [5] Rakesh Ranjan Kumar, **Siba Mishra**, and Chiranjeev Kumar. Nov. 2017. "Prioritizing the solution of cloud service selection using integrated MCDM methods under Fuzzy environment". In: *The Journal of Supercomputing* 73.11, pp. 4652–4682.
- 2016 [6] **Siba Mishra** and Chiranjeev Kumar. 2016. "A novel adaptive structure for SOA system effort estimation". In: *Transactions on Emerging Telecommunications Technologies* 27.8, pp. 1115–1127.
- 2013 [7] Abhishek Ray, **Siba Mishra**, and Durga Prasad Mohapatra. Jan. 2013. "A Novel Approach for Computing Dynamic Slices of Concurrent Aspect-Oriented Programs". In: *International Journal of Software Engineering and its Applications (IJSEIA)* 7.1, pp. 13–32.
- 2012 [8] Abhishek Ray, **Siba Mishra**, and Durga Prasad Mohapatra. Sept. 2012. "A Novel Approach for Computing Dynamic Slices of Aspect-Oriented Programs". In: *International Journal of Computer Information Systems (JCIS)* 5.3, pp. 6–12.

Publications (Conference Proceedings)

- 2026 [1] Unnati Shah and **Siba Mishra**. May 2026. "Operationalization-Based Modeling of Software Non-Functional Requirement Relationships: A Context-Aware Approach". In: *The International FLAIRS Conference Proceedings. [FLAIRS'2026]*, Florida, USA 39.1. doi: 10.32473/flairs.39.1.141431. URL: <https://journals.flvc.org/FLAIRS/article/view/141431>.
- 2025 [2] Kritika Das, Pranab Kumar Sahoo, Kumar Sumit Sohan, Supriya Panigrahy, and **Siba Mishra**. Apr. 2025. "Exploring Radial Basis Function Neural Networks with Varied Loss Functions". In: *Fourth International Conference on Computing, Communication and Learning [COCOLE'2025]*, NIT Rourkela, pp. 429–443.
- [3] Sanjaya Kumar Panda, Sankalp Dubey, and **Siba Mishra**. Oct. 2025. "Efficient Task Scheduling Algorithms for Decentralized Large Language Model Serving". In: *IEEE Region 10 Conference (TENCON), [TENCON'2025]*, Kota Kinabalu, Sabah, Malaysia, pp. 1023–1027.
- [4] Kusagra Raj, Himanshu, Mrinal Das, Abhishek Kumar Sharma, Supriya Panigrahy, and **Siba Mishra**. Apr. 2025. "Enhancing Coverless Image Steganography with Combined SIFT and ORB Features". In: *Fourth International Conference on Computing, Communication and Learning [COCOLE'2025]*, NIT Rourkela, pp. 398–407.
- [5] Sai Sipraranjan, Pragyna Mahapatra, **Siba Mishra**, Supriya Panigrahy, Chinmaya Jena, Raj Kumar Rout, Sandip Kumar Sahoo, Bijaya Laxmi Swain, and Rajashree Sahu. Aug. 2025. "Skin Cancer Image Classification Using Convolutional Neural Networks". In: *International Conference on Artificial intelligence and Emerging Technologies [ICAET'2025]*, XIM University, Bhubaneswar.
- [6] Bijaya Laxmi Swain, Sandip Kumar Sahoo, **Siba Mishra**, Supriya Panigrahy, Rajshree Sahu, Raj Kumar Rout, Chinmaya Jena, Sai SipraRanjan, and Pragyna Mahapatra. Aug. 2025. "Early Detection of Diabetic Retinopathy Using Convolutional Neural Networks". In: *2025 International Conference on Artificial intelligence and Emerging Technologies [ICAET'2025]*, XIM University, Bhubaneswar.

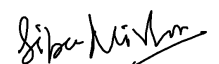
- 2021 [7] **Siba Mishra** and Arpit Sharma. Feb. 2021. "A Generalized Semantic Filter for Glossary Term Extraction from Large-Sized Software Requirements". In: *Proceedings of the 14th Innovations in Software Engineering Conference (Formerly Known as India Software Engineering Conference) [ISEC'2021]*, Bhubaneswar, India.
- [8] **Siba Mishra** and Arpit Sharma. Feb. 2021. "Crawling Wikipedia Pages to Train Word Embeddings Model for Software Engineering Domain". In: *Proceedings of the 14th Innovations in Software Engineering Conference (Formerly Known as India Software Engineering Conference) [ISEC'2021]*, Bhubaneswar, India.
- 2020 [9] Kushagra Bhatia, **Siba Mishra**, and Arpit Sharma. Feb. 2020. "Clustering Glossary Terms Extracted from Large-Sized Software Requirements Using FastText". In: *Proceedings of the 13th Innovations in Software Engineering Conference (Formerly Known as India Software Engineering Conference) [ISEC'2020]*, Jabalpur, India.
- [10] **Siba Mishra** and Arpit Sharma. Mar. 2020. "Automatic Word Embeddings-Based Glossary Term Extraction from Large-Sized Software Requirements". In: *Proceedings of the 26th International Working Conference on Requirements Engineering: Foundation for Software Quality [REFSQ'2020]*, Pisa, Italy, pp. 203–218.
- 2019 [11] **Siba Mishra** and Arpit Sharma. Sept. 2019. "On the Use of Word Embeddings for Identifying Domain Specific Ambiguities in Requirements". In: *Proceedings of the 27th IEEE International Requirements Engineering Conference Workshops [REW'2019]*, Jeju Island, South Korea, pp. 234–240.
- 2017 [12] **Siba Mishra** and Chiranjeev Kumar. Sept. 2017. "Improving the trust of end users in enterprise SOA using combinatorial group testing methods". In: *Proceedings of the 12th International Conference on Digital Information Management [ICDIM'2017]*, Fukuoka, Japan, pp. 144–149.
- 2014 [13] **Siba Mishra** and Chiranjeev Kumar. Nov. 2014. "Estimating development size and effort of business process service-oriented architecture applications". In: *Proceedings of the 2nd International Conference on Systems and Informatics [ICSAI'2014]*, Shanghai, China, pp. 1006–1011.
- [14] **Siba Mishra**, Urvashi Sharma, and Chiranjeev Kumar. Feb. 2014. "A Novel Approach for Computing Dynamic Slices of Web Based Applications". In: *Proceedings of the Fourth International Conference on Advanced Computing Communication Technologies [ACCT'2014]*, Rohtak, India, pp. 153–158.
- 2013 [15] Abhishek Ray, **Siba Mishra**, and Durga Prasad Mohapatra. Feb. 2013. "Architectural Aspect-Oriented Dynamic Slicing". In: *Proceedings of the Fourth Workshop on Advances in Model-Based Software Engineering [WAMBSE'2013]*, co-located at *[ISEC'2013]*, New Delhi, India. Vol. 11. Infosys Labs Briefings.

Manuscripts (Accepted Without Issue or Under Review)

- Submitted on 22nd June, 2026** Unnati Shah and **Siba Mishra**. "Adaptive QARM: LLM-Driven Context-Aware NFR Conflict Detection for Software Reliability". The 37th IEEE International Symposium on Software Reliability Engineering (ISSRE'26), October 20-23, 2026. Limassol, Cyprus. **(Accepted on 04-08-2026)**.
- Submitted on 27th June, 2026** **Siba Mishra**, Unnati Shah, Vishruti V Desai. "ModernBERT for Software Engineering: Domain Adaptation, Contextual Embedding, and Empirical Trade-off Analysis". 20th Innovations in Software Engineering Conference (ISEC'27), 18-20 February, 2027. Mumbai, India. **(Under Review)**.
- Submitted on 12th August, 2026** Rahul Das, Dillip Rout, **Siba Mishra**, Akashdeep Baruah and Deepshikha Routray. "Combining Undersampling and Oversampling to Boost Sentiment Classification of Reviews on Software Engineering Domain Courses from Coursera Repository". Knowledge and Information Systems (KAIS) Journal. **(Under Review)**.
- Submitted on 25th August, 2026** Rakesh Ranjan Kumar, **Siba Mishra**, Binita Kumari, Debendra Muduli, Nand Kumar Jyotish, Chiranjeev Kumar. "MOSAIC: Multi-Agent Orchestrated System for Automated Intelligent Cloud Service Selection". Future Generation Computer Systems (FGCS) Journal. **(Under Review)**.

Declaration

I hereby declare that the above-mentioned information is correct up to my knowledge, and I bear the responsibility for the correctness of the above-mentioned particulars.



Siba Mishra, PhD